

GEOL 211 • Exam 2 Study Guide

Atmospheric Circulation + Ocean Circulation · built from the class slides + review guide

Exam: Wed 9/30 · on paper · individual part (30 min) FIRST, then team part (20 min) · questions also come from **class assignments**, so add them to your notes

How to copy this into your notebook: (1) Copy **The Big Picture** below first — it's the map of the whole unit. (2) For each topic, copy the **question**, the **ANSWER box**, the **numbered steps** and the **diagram** (that's what answers "explain why" questions). "Know these" + exam traps come next. (3) Fill in the page numbers in the contents table as you go — that's your index during the exam. Short on time? Do the ★ topics first.

The Big Picture — how everything connects

This whole unit is **one chain of cause → effect**. Every topic is one link in it:

1. The SUN heats Earth unevenly — hot at the equator, cold at the poles (topic 2)



2. Heat changes air density — warm/humid air is light and RISES (= LOW pressure, rain); cold/dry air is heavy and SINKS (= HIGH pressure, dry) (topics 1–2)



3. Air flows from HIGH to LOW = WIND (topic 2)



4. Earth's spin bends the wind (CORIOLIS) — right in the Northern Hemisphere, left in the Southern (topic 4)



5. Result: global wind belts — trade winds, westerlies, polar easterlies (3 cells per hemisphere) (topic 3)

- Land heats/cools faster than water → sea breezes + monsoons (topic 5)
- Warm ocean + moist air + calm upper winds → HURRICANES, steered around highs (topics 6–9)



6. Wind drags the ocean surface → SURFACE CURRENTS + GYRES (bent by Coriolis again) (topics 11–14)

- Wind pushes surface water away → deep water comes up = UPWELLING (nutrients → fish) (topic 15)
- Trade winds weaken or strengthen → EL NIÑO / LA NIÑA (topic 16)



7. Separately, DENSITY drives the deep ocean — cold + salty water sinks near the poles → slow global CONVEYOR BELT that moves heat (~1000 yrs per loop) (topics 10, 12, 17)

3 rules that answer half the exam

1. **Warm or humid = light = rises = LOW = rain.** Cold or dry = heavy = sinks = HIGH = dry.
2. **Air and water move from HIGH → LOW** and get bent **RIGHT in the NH, LEFT in the SH** (Coriolis).
3. **Land heats up AND cools down faster than water.**

Abbreviations: NH / SH = Northern / Southern Hemisphere · H / L = high / low pressure · SST = sea-surface temperature · ITCZ = Intertropical Convergence Zone · ↑ = increases · ↓ = decreases

Quick Answers — fastest lookup during the exam

If the question is about...	Remember...
Air density	Warm or humid air = LESS dense → rises. Cold or dry air = MORE dense → sinks. Densest air = poles.
Low vs high pressure	LOW = rising air → clouds, rain. HIGH = sinking air → clear, dry. Wind blows H → L.
Pressure belts	0° LOW (ITCZ, wet) · 30° HIGH (dry, deserts) · 60° LOW (wet) · 90° HIGH (cold, dry)
Cells + winds	Hadley 0–30° (trade winds) · Ferrel 30–60° (westerlies) · Polar 60–90° (polar easterlies)
Coriolis	Bends RIGHT in NH, LEFT in SH · zero at equator, stronger toward poles · doesn't cause wind
Breezes	DAY = sea breeze (sea → land) · NIGHT = land breeze (land → sea)
Monsoons	SUMMER: ocean → land = WET · WINTER: land → ocean = DRY · follows the ITCZ
Hurricane needs	Water >27°C · low pressure · warm, moist air · wind shear <20 knots · Coriolis (not at equator)
Hurricane fuel	LATENT HEAT released when water vapor condenses · spins counterclockwise in NH
Hurricane hazards	Storm surge (worst) · wind · rain · flooding · waves · landslides
Hurricane path	Goes AROUND highs (clockwise in NH): west on trades → north → northeast on westerlies
Climate change	Not MORE hurricanes — STRONGER, WETTER, higher surge
Surface vs deep currents	Surface = WIND, ~10% of ocean, fast · Deep = DENSITY, ~90%, slow
Thermocline	Temp drops fast: 200 m (13°C) → 1000 m (4°C) · barrier to mixing
Ekman transport	Surface water 45° right of wind · NET transport 90° RIGHT of wind (NH), LEFT (SH)
Gyres	CLOCKWISE in NH, COUNTERCLOCKWISE in SH · warm current on WEST side, cold on EAST side
West vs east boundary	West (Gulf Stream) = fast, narrow, deep, warm · East (Canary) = slow, wide, shallow, cold
Upwelling	Cold, nutrient-rich water rises → plankton → fish · NH west coast + wind from the NORTH
El Niño	Trades WEAKEN → warm water moves east → no upwelling off Peru → FEWER Atlantic hurricanes
La Niña	Trades STRONGER → colder east Pacific → MORE Atlantic hurricanes
Conveyor belt	Sinks near Greenland + Antarctica · ~1000 yrs · moves heat · fresh meltwater SLOWS it

Contents — write your notebook page numbers here

PART A • ATMOSPHERE			PART B • OCEAN		
1	Air density — how + why it changes	p. ____	10	Surface vs deep water ★	p. ____
2	High + low pressure — why + weather	p. ____	11	Major ocean currents (gyres)	p. ____
3	Global atmospheric circulation ★	p. ____	12	What drives surface + deep currents ★	p. ____
4	Coriolis effect ★	p. ____	13	West vs east boundary currents ★	p. ____
5	Sea/land breezes + monsoons	p. ____	14	How currents affect weather	p. ____
6	What a hurricane needs ★	p. ____	15	Upwelling + downwelling ★	p. ____
7	Hurricane hazards + storm surge	p. ____	16	El Niño + La Niña ★	p. ____
8	Predicting a hurricane's path ★	p. ____	17	Deep water circulation + climate ★	p. ____
9	Hurricanes + climate change	p. ____			

PART A • THE ATMOSPHERE (topics 1–9)

1 How and why does air density change?

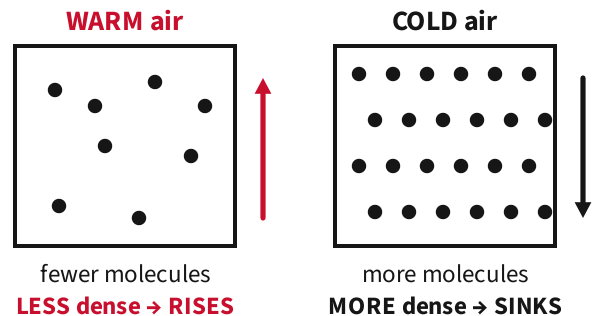
Review guide: Explain how/why air density changes.

ANSWER — copy this

Air density depends on **temperature** and **moisture**. **Warm air and humid air are LESS dense, so they rise.** Cold air and dry air are MORE dense, so they sink.

HOW IT WORKS

1. **Warm air:** molecules move fast and spread out → fewer molecules in the same space → less dense → **rises**.
2. **Cold air:** molecules slow down and pack together → more molecules in the same space → denser → **sinks**.
3. **Humid air:** a water vapor molecule (H_2O , mass 18) is **lighter** than the N_2 (28) and O_2 (32) it replaces → **humid air is LESS dense than dry air**.



How to read it: same-size boxes. Fewer molecules = lighter = the air rises.

KNOW THESE

- Air moves 2 ways: **horizontally** (controlled by **pressure**, high → low) and **vertically** (controlled by **density**).
- Densest air on Earth: at the **poles** (cold + dry).
- Air = 78% N_2 , 21% O_2 , 0–4% water vapor.
- The **Sun** is the energy source for all wind and ocean circulation.

★ **Exam trap:** "Humid air is heavier" is **false** — humid air is **LIGHTER** than dry air.

2 Why do high and low pressure form, and what weather do they bring?

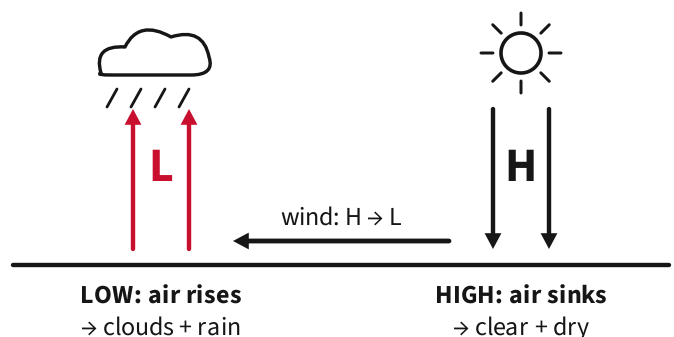
Review guide: Explain why regions of high/low density develop. Identify high/low pressure. Describe the weather.

ANSWER — copy this

The Sun heats Earth **unevenly**. Where air is warmed it **rises**, leaving **LOW** pressure at the surface (**clouds + rain**). Where air is cooled it **sinks** and piles up as **HIGH** pressure (**clear + dry**). Air then flows from high to low = **wind**.

HOW IT WORKS

1. **Equator:** Sun is straight overhead → energy concentrated → hot. **Poles:** Sun is low → the same sunbeam spreads over more area → cold.
2. So there's a heat **surplus** near the equator and a **deficit** near the poles → air + oceans carry heat toward the poles.
3. **Rising** air expands and cools → water vapor condenses → **clouds, rain**.
4. **Sinking** air compresses and warms → no condensation → **clear skies**.



How to read it: air rises at the L and sinks at the H; at the ground the wind blows from H to L.

	LOW pressure (L)	HIGH pressure (H)
Air	warm, humid, RISES	cold, dry, SINKS
Weather	cloudy, rainy, stormy	clear, dry, calm
Surface winds	flow IN toward it	flow OUT away from it
Spin (NH)	counterclockwise	clockwise
Where	0° (ITCZ) and 60°	30° and 90° (poles)

KNOW THESE

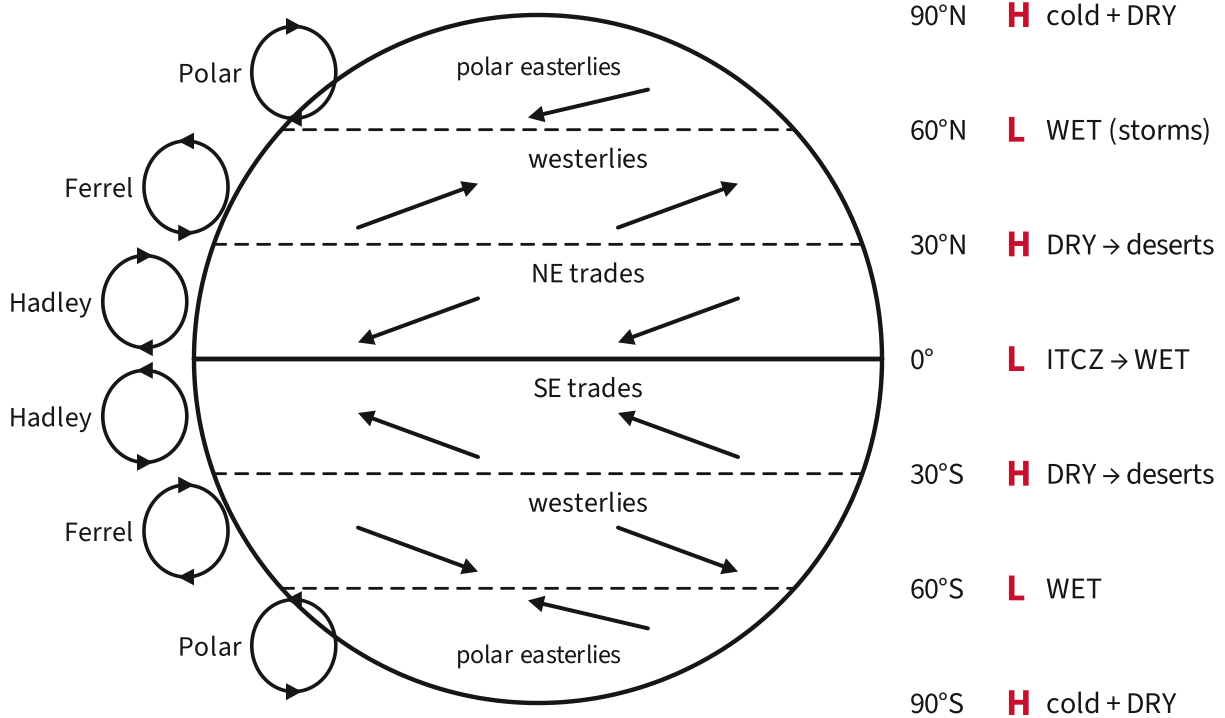
- Seasons shift where the Sun is most direct, so the belts shift north and south during the year.

3 Draw the global atmospheric circulation pattern ★

Review guide: Draw the global circulation. Label high/low pressure, the ITCZ, and high/low precipitation.

ANSWER – copy this

Each hemisphere has **3 cells**: Hadley (0–30°), Ferrel (30–60°), Polar (60–90°). Air **RISES at 0° and 60°** (LOW → wet) and **SINKS at 30° and 90°** (HIGH → dry). Surface winds blow from the highs to the lows and get bent by Coriolis → **trade winds, westerlies, polar easterlies**.



How to read it: the loops on the left are side views of the cells (air going up and down). The arrows on the globe are the surface winds. Every arrow starts at a H line and points toward a L line, bent right in the NH and left in the SH.

HOW TO DRAW IT IN 6 STEPS

1. Draw a circle. Add lines at 0° , 30°N , 60°N , 30°S , 60°S .
2. Write the pressure pattern going out from the equator: **L - H - L - H** (0° L , 30° H , 60° L , 90° H), same in both hemispheres.
3. Label rain: $0^\circ = \text{ITCZ}$, WET · $30^\circ = \text{DRY}$ (deserts) · $60^\circ = \text{WET}$ · $90^\circ = \text{DRY}$ (cold desert).
4. Draw surface winds from each H toward the next L: $30^\circ \rightarrow 0^\circ$ (trades), $30^\circ \rightarrow 60^\circ$ (westerlies), $90^\circ \rightarrow 60^\circ$ (polar easterlies).
5. Bend every arrow: **right in the NH, left in the SH**. That gives NE trades, SE trades, westerlies, polar easterlies.
6. Draw the loops on the side and name them: Hadley ($0-30^\circ$), Ferrel ($30-60^\circ$), Polar ($60-90^\circ$).

KNOW THESE

- **ITCZ** (Intertropical Convergence Zone, the "doldrums") = where NE + SE trades meet at the equator → rising air → heavy rain. It follows the Sun: farther **north in July, south in January**.
- **30° = horse latitudes / subtropical highs** → the world's deserts (Sahara, Arabian, Kalahari, Australian).
- **60° = subpolar lows** → rain, snow, storms. **90° = polar highs** → cold and dry.
- Winds are named for where they come **FROM** (NE trades blow from the northeast).
- A **non-rotating** Earth would have just 1 big cell per hemisphere. Rotation breaks it into 3.

4

Why does the Coriolis effect exist, and what does it do? ★

Review guide: Explain why the Coriolis Force exists and what the result is.

ANSWER — copy this

Earth spins west → east, and places near the equator move **faster** than places near the poles. Anything moving north or south keeps its old eastward speed, so it drifts off course. **Result: moving objects bend RIGHT in the NH and LEFT in the SH.**

HOW IT WORKS

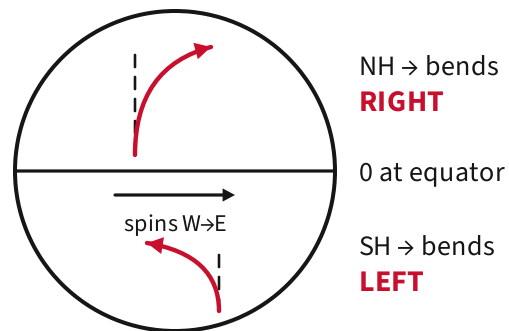
1. Every point on Earth makes one full turn per day.
2. The equator has the longest trip (~40,000 km/day) so it's fastest. 60° goes ~20,000 km/day. The poles barely move.
3. Air moving from the equator toward the North Pole is still moving east faster than the ground under it → it gets ahead → curves to the **right**.

KNOW THESE

- It's an **apparent** deflection — we see it because we're standing on a spinning planet.
- **Zero at the equator**, gets stronger toward the poles.
- Stronger for faster, heavier objects. Only matters for big motions (tens of km or more).
- It does **NOT cause** wind — it only changes the wind's **direction**.

WHAT IT CAUSES

- Bends the global winds into trade winds, westerlies and polar easterlies (topic 3).
- Makes ocean gyres turn and causes Ekman transport (topics 11–12).
- Makes storms spin: in the NH, air flowing into a LOW veers right → **counterclockwise**. That's why hurricanes can't form right at the equator.



dashed = path with no spin

How to read it: dashed = path with no spin. Both objects head north: the NH one bends right, the SH one bends left.

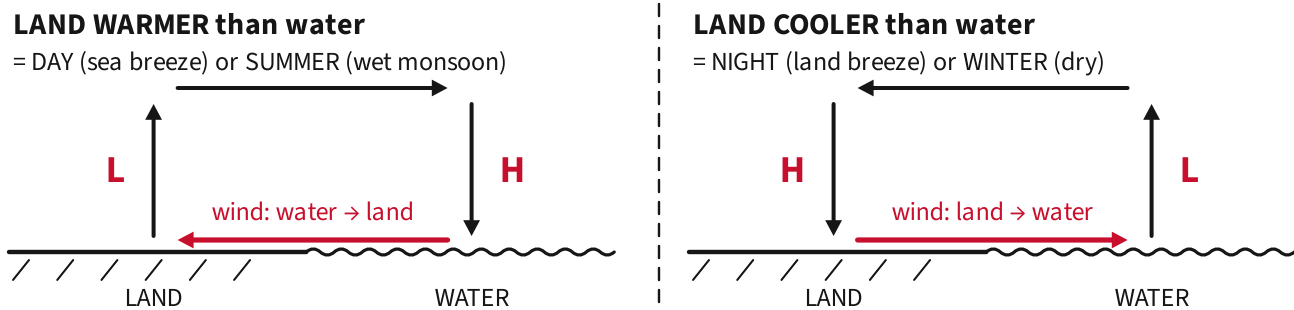
★ **Exam trap:** "Coriolis causes wind" is **false**. Pressure differences cause wind; Coriolis just bends it.

5 Explain sea/land breezes and monsoons

Review guide: Explain the development of monsoons + land/sea breeze.

ANSWER — copy this

Both happen because **LAND heats up and cools down FASTER than WATER** (water has a high heat capacity). Air rises over whichever surface is **warmer** (LOW) and sinks over the **cooler** one (HIGH), so the wind blows from the cool surface toward the warm one. **Breezes flip every day; monsoons flip every season.**



How to read it: left = land is warmer (daytime or summer) → wind blows from the water onto the land. Right = land is cooler (night or winter) → the whole loop reverses.

When	Warmer surface (L)	Surface wind blows	Result
DAY	land	sea → land (onshore)	SEA breeze
NIGHT	sea	land → sea (offshore)	LAND breeze
SUMMER	land	ocean → land (moist air rises, cools)	WET monsoon — heavy rain
WINTER	ocean	land → ocean (dry air)	DRY season

KNOW THESE

- A breeze is named for where it comes **FROM** (sea breeze comes from the sea).
- Monsoon = a **seasonal reversal of winds**. The biggest is India / South Asia.
- Monsoons follow the **ITCZ**: June–July it moves north over India → rain there (Kozhikode). December–January it moves south → rain at Darwin, Australia.

6 What conditions does a major hurricane need? ★

Review guide: List and describe the conditions necessary for a major hurricane.

ANSWER — copy this

Class list — memorize all 5: **(1) warm water >27°C, (2) a low-pressure system, (3) warm air with lots of water vapor, (4) low wind shear (<20 knots), (5) strong enough Coriolis** (so not at the equator).

Condition	Why it matters
1. Warm water >27°C	supplies heat + moisture = the fuel
2. Low-pressure system	the starting point: air flows in and rises
3. Warm air, high water vapor	more vapor = more latent heat released when it condenses
4. Low wind shear (<20 knots)	winds at all heights similar → storm stays upright. High shear tilts it and tears it apart
5. Strong Coriolis	makes it spin → no hurricanes right at the equator

THE ENGINE = LATENT HEAT

1. Warm ocean water evaporates (evaporation **absorbs** heat).
2. Moist air spirals in toward the low and rises in the eyewall.
3. The water vapor condenses into clouds → **releases** that heat into the storm.
4. The air warms and rises faster → pressure drops more → winds speed up → the cycle repeats.

KNOW THESE

- Air rushes toward the low and veers right → spins **counterclockwise in the NH** (clockwise in the SH).
- Structure: calm **eye** (~13–16 km wide) · **eyewall** = strongest winds (can top 250 km/h) · rain bands spiral out.
- Form over warm tropical water (none in the S Atlantic). Die over land or cold water (no fuel) or in high shear.
- Hurricane (Atlantic) = typhoon (W Pacific) = tropical cyclone — same storm.

7 What are the risks (hazards) of hurricanes?

Review guide: List and describe the risks associated with hurricanes.

ANSWER — copy this

Six hazards: **waves, wind, rain, flooding, landslides, storm surge**. **Storm surge** — ocean water pushed up onto the land — is usually the most destructive.

STORM SURGE EXPLAINED

1. Mostly caused by strong **onshore winds** pushing water toward land (**wind-driven**).
2. The storm's low pressure lifts the sea only a little (about **5%** of the surge).
3. As the water reaches shallow water near the coast it **piles up**.
4. Worst if it arrives at **high tide** (Hurricane Sandy 2012, Atlantic City tide gauge).

Factor	Surge is HIGHER when...
Wind speed	winds are stronger
Size (radius) of storm	the storm is bigger
Forward speed	fast storm → higher surge on the open coast · slow storm → floods longer and farther inland
Angle of approach	it hits the coast head-on (perpendicular) instead of sliding along it (parallel)
Continental shelf	the shelf is wide and gently sloping (shallow water)
Shape of coastline	bays and inlets funnel the water in

KNOW THESE

- Rain flooding can hit far inland — e.g., Hurricane Ivan flooded Asheville, NC (2004).

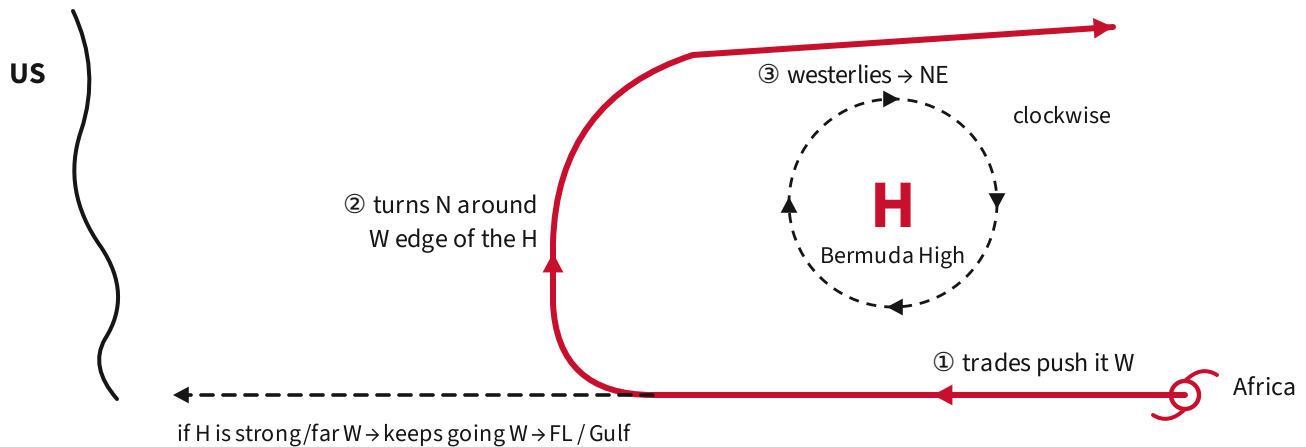
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Predict a hurricane's path from the highs and lows ★

Review guide: Predict the path of a hurricane based on high/low pressure.

ANSWER — copy this

Hurricanes are **steered by the winds around pressure systems**. They can't push through a HIGH, so they **travel around its edge — clockwise around a high in the NH** — and they get pulled toward LOWS.



How to read it: the storm (red) rides the clockwise flow around the Bermuda High: west, then north, then northeast.

TYPICAL ATLANTIC PATH

1. Forms off Africa → the **trade winds** push it **WEST**.
2. Reaches the west edge of the **Bermuda High** → turns **NORTH**.
3. Meets the **westerlies** (~30°N) → curves **NORTHEAST**, out to sea.

RULES FOR PREDICTING

- High is strong / stretches far west → storm keeps going west → Florida / Gulf of Mexico.
- High is weak or shifted east → storm turns north early → out to sea or up the East Coast.
- A low (trough) over the eastern US pulls the storm north toward the coast.

★ **Exam trap:** On a map question: find the H, sketch clockwise arrows around it, then slide the storm along the H's edge toward the nearest L.

9

How will hurricanes change as the climate warms?

Review guide: Explain expected changes in hurricane activity due to climate change + trends over the last 100 years.

ANSWER — copy this

Expect **not MORE hurricanes, but STRONGER and WETTER ones**, with higher storm surge.

EXPECTED (FUTURE)

- Warmer ocean = more fuel → **stronger** storms (more Cat 4–5) that intensify faster.
- Warmer air holds more water vapor → **more rain** + flooding.
- Sea-level rise → **higher storm surge**, more coast flooded.
- Total number of storms: about the same or fewer.

OBSERVED (LAST ~100 YEARS)

- Total number: **no clear long-term trend** (old records missed storms at sea — no satellites before the 1960s).
- The **share of major storms** (Cat 3–5) and their rainfall has gone up.
- Damage (\$) has gone up a lot — mostly because more people and buildings are on the coast.

PART B • OCEAN CIRCULATION (topics 10–17)

Why ocean currents matter: they regulate **climate** (move and store heat, store gases) and control **marine productivity** (supply nutrients and oxygen). Ocean circulation is controlled by **wind, density, the position of the continents, and Coriolis**.

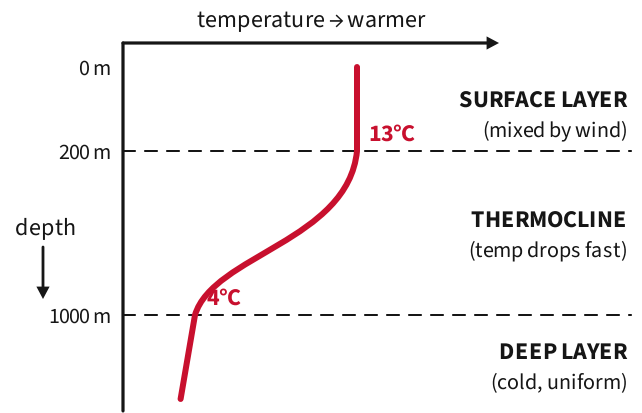
10 Surface water vs deep water ★

Review guide: Describe the properties of surface and deep water.

ANSWER — copy this

The ocean is **layered by density**. A warm, light, wind-mixed **surface layer** sits on top of a cold, dense, dark **deep layer**. In between is the **THERMOCLINE**, where temperature drops fast ($\approx 13^{\circ}\text{C}$ at 200 m $\rightarrow \approx 4^{\circ}\text{C}$ at 1000 m). It acts like a barrier, so the two layers barely mix.

	SURFACE layer	DEEP layer
Temperature	warm (varies)	cold ($\approx 4^{\circ}\text{C}$ or less)
Density	low	high
Light	sunlit $\rightarrow$ photosynthesis	dark
Nutrients	LOW (plankton use them up)	HIGH (from decaying material)
Oxygen	high (air + photosynthesis)	lower
Mixing	mixed by wind + waves	not mixed by wind
Moved by	wind $\cdot$ $\sim 10\%$ of ocean $\cdot$ fast	density $\cdot$ $\sim 90\%$ $\cdot$ slow



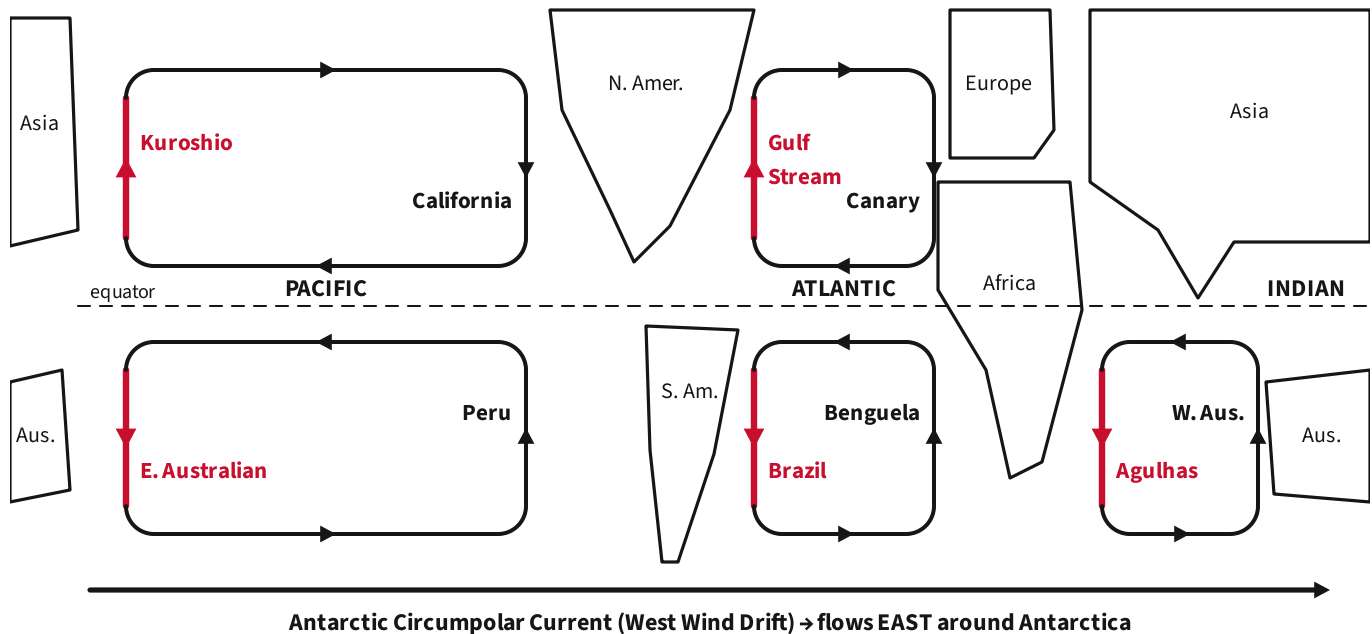
How to read it: red line = temperature going down. It stays warm in the mixed layer, drops fast in the thermocline, then stays cold.

SALINITY (SALTINESS)

- Measured in parts per thousand (ppt, ‰) — average seawater $\approx$ **35 ppt** (34.7).
- Salt comes from erosion of the continents + hydrothermal vents.
- **Highest** in the subtropics ($23.5\text{--}35^{\circ}$): evaporation $>$ rain under the dry 30° highs. The Atlantic is the saltiest ocean.
- **Lowest** at the equator (lots of rain) and near big rivers (Amazon, Ganges).

ANSWER — copy this

Surface currents form giant loops called **GYRES** — **clockwise in the NH, counterclockwise in the SH**. Every gyre has a **WARM current on its WEST side** (flowing toward the pole) and a **COLD current on its EAST side** (flowing toward the equator).



How to read it: simplified world map (Pacific on the left, like the class map). Red = warm west-side current, black = cold east-side current. Arrows show the spin direction.

Gyre	WEST side — warm	EAST side — cold
North Atlantic	Gulf Stream	Canary
South Atlantic	Brazil	Benguela
North Pacific	Kuroshio	California
South Pacific	East Australian	Peru (Humboldt)
Indian	Agulhas	West Australian

ALSO KNOW

- **North Atlantic Current** carries Gulf Stream water to Europe. **North Pacific Current** crosses the top of the N Pacific gyre.
- **North + South Equatorial Currents** flow west (pushed by the trades). The **Equatorial Countercurrent** flows east between them.
- **Antarctic Circumpolar Current** (West Wind Drift) flows east all the way around Antarctica — the biggest current.
- Cold currents coming down from the north: **Labrador** (off Canada), **Oyashio** (off Japan).
- Gulf Stream: first seen 1513 (Ponce de León), charted by Ben Franklin (1770s); flows ~300× more water than the Amazon.

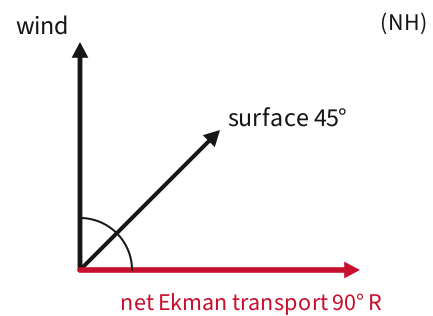
ANSWER — copy this

Surface currents are driven by **WIND**, then turned by Coriolis and blocked by continents. **Deep currents** are driven by **DENSITY** differences — cold, salty water is heavy, so gravity pulls it down (called **thermohaline** circulation: thermo = temperature, haline = salt).

	SURFACE currents	DEEP currents (thermohaline)
Driven by	wind	density (temperature + salt) → gravity
Share of ocean	~10%	~90%
Speed	fast	slow
Where	above the thermocline	below the thermocline

HOW WIND MOVES WATER: EKMAN TRANSPORT

1. Wind drags the top layer of water.
2. Coriolis turns it: surface water moves **45° to the right** of the wind (NH).
3. Each layer drags the one below it, a bit slower and turned a bit more → the **Ekman spiral**.
4. The **net** movement of all that water = **90° to the RIGHT of the wind** in the NH (90° LEFT in the SH). This is **Ekman transport**.



How to read it: wind blows north; the water ends up moving east (90° right).

HOW DENSITY IS SET (DEEP CURRENTS)

- Density is "fixed" at the **surface** — then the water sinks and keeps that density.
- **Colder = denser** (water is cooled by contact with the cold air).
- **Saltier = denser**. Salt goes UP with evaporation and **sea-ice formation** (freezing leaves the salt behind). Salt goes DOWN with rain, rivers and melting ice.
- The coldest, saltiest water forms near the **poles** → that's where deep water sinks.

★ **Exam trap:** Ekman transport is **90°**, not 45°. 45° is only the very top layer.

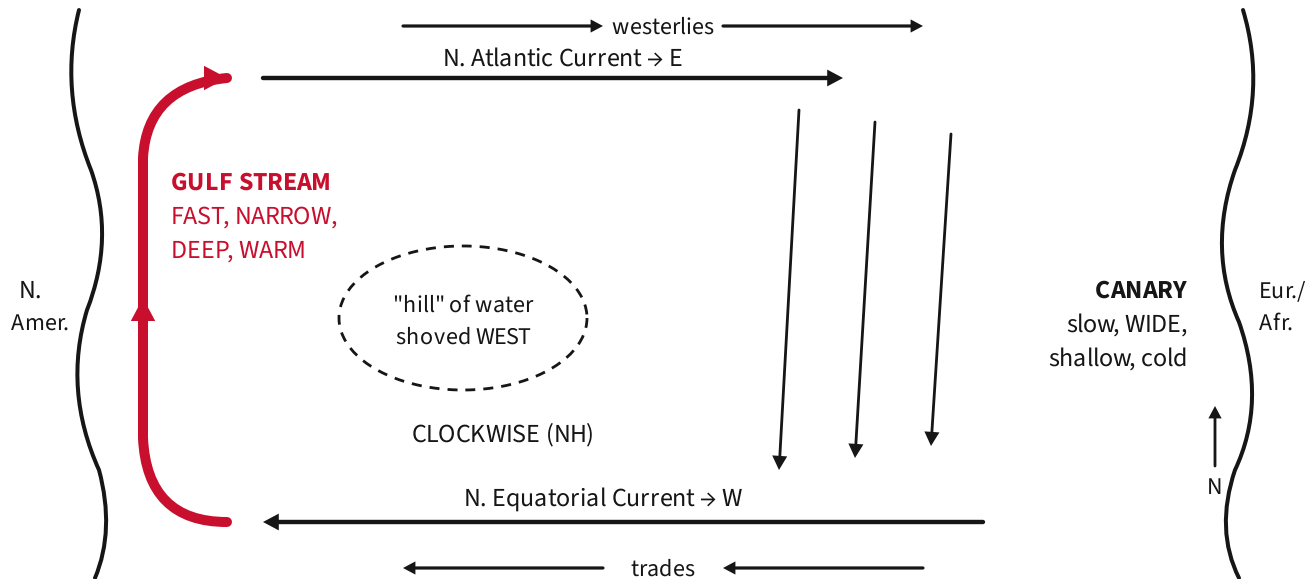
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Why are western boundary currents fast and eastern ones slow? ★

Review guide: Explain why the speed of surface currents varies (eastern/western boundary currents).

ANSWER – copy this

Currents on the **west side** of an ocean (Gulf Stream) are **fast, narrow, deep and warm**. Currents on the **east side** (Canary) are slow, wide, shallow and cold. This is **western intensification**, caused by **Earth rotating eastward** and **Coriolis getting stronger away from the equator**. Together they squeeze the gyre's flow against the west side, so the same water has to move faster through a narrow path.



How to read it: one NH gyre (North Atlantic). The "hill" of water in the middle is shoved west, so the Gulf Stream is squeezed into a thin, fast current, while the Canary Current spreads out wide and slow.

	WESTERN boundary	EASTERN boundary
Speed	FAST	slow
Width	narrow	wide
Depth	deep	shallow
Temperature	warm (comes from the equator)	cold (comes from high latitudes)
Flow	Gulf Stream ~55 Sv	Canary ~16 Sv
Examples	Gulf Stream, Kuroshio	Canary, California

KNOW THESE

- Sv = Sverdrup = 1 million m³ of water per second.
- Think of a river: where it's narrow the water runs fast; where it's wide it runs slow.

14

How do surface currents affect weather?

Review guide: Explain how surface currents affect weather.

ANSWER – copy this

Currents move **heat** from the equator toward the poles. **Warm currents** warm the air and add moisture → warm, humid, rainy coasts. **Cold currents** cool the air and add little moisture → cool, foggy, dry coasts (even deserts).

	WARM current	COLD current
Where	west side of a gyre → east coasts of continents	east side of a gyre → west coasts
Effect on air	warm + humid, lots of evaporation	cool, little evaporation, fog
Weather	rain; fuels storms + hurricanes	dry — coastal deserts
Examples	Gulf Stream → humid US East Coast	California Current → cool, foggy California; Peru + Benguela → Atacama + Namib deserts

CLASS EXAMPLE

- The Gulf Stream + North Atlantic Current keep Western Europe mild. January average high/low: **Newfoundland 30/16°F vs Ireland 46/37°F** — about the same latitude!
- Water's high heat capacity → coastal climates are milder than inland ones.

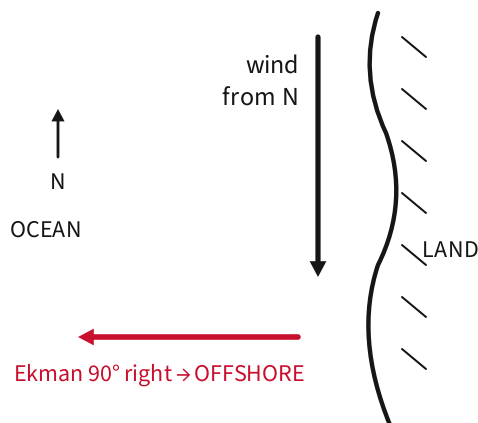
15 How do deep and surface water mix? (upwelling + downwelling) ★

Review guide: Explain upwelling/downwelling. What does it mean for marine life and climate?

ANSWER — copy this

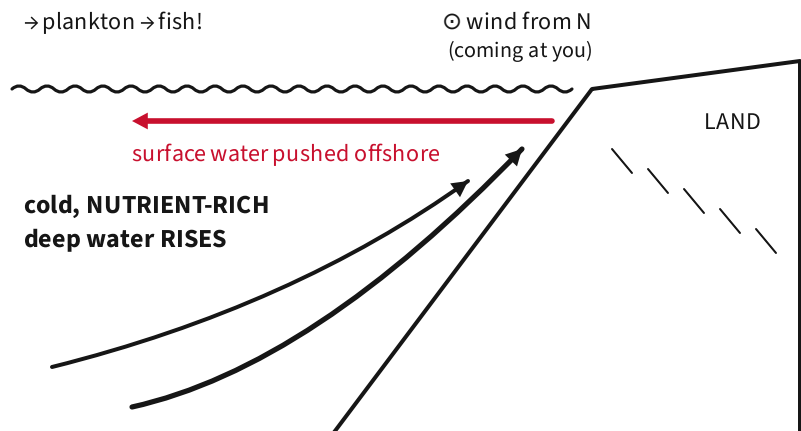
UPWELLING = cold, nutrient-rich deep water rises to the surface. **DOWNWELLING** = surface water piles up and sinks. Both are mostly caused by **wind + Ekman transport** moving surface water **away** from an area (upwelling) or **toward** it (downwelling).

MAP VIEW (NH west coast)



SIDE VIEW

→ plankton → fish!



How to read it: left = map view of a NH west coast (like California). Wind blows south along the coast; Ekman transport moves water 90° right = offshore. Right = side view: deep water rises to replace it.

Where	What happens	Why
NH west coast, wind from the north (California)	UPWELLING	Ekman moves water offshore → deep water rises to replace it
NH west coast, wind from the south	DOWNWELLING	Ekman moves water onshore → it piles up at the coast and sinks
Equator	UPWELLING	Coriolis turns water right (north of equator) and left (south) → water spreads apart
Around Antarctica	UPWELLING	winds push surface water apart
Center of a gyre	DOWNWELLING	water converges and piles up

WHY IT MATTERS: MARINE LIFE ★

- Upwelling brings nutrients up into the sunlight → phytoplankton bloom → zooplankton → **fish**.
- Best fisheries: California, Peru, Canary, Benguela — all **eastern boundary currents**.
- Downwelling carries **oxygen** down to deep-sea life, but leaves the surface nutrient-poor → low productivity.

WHY IT MATTERS: CLIMATE

- Upwelled cold water cools the coast → cool, foggy weather.
- Mixing moves heat and gases (O₂, CO₂) between the surface and the deep ocean.

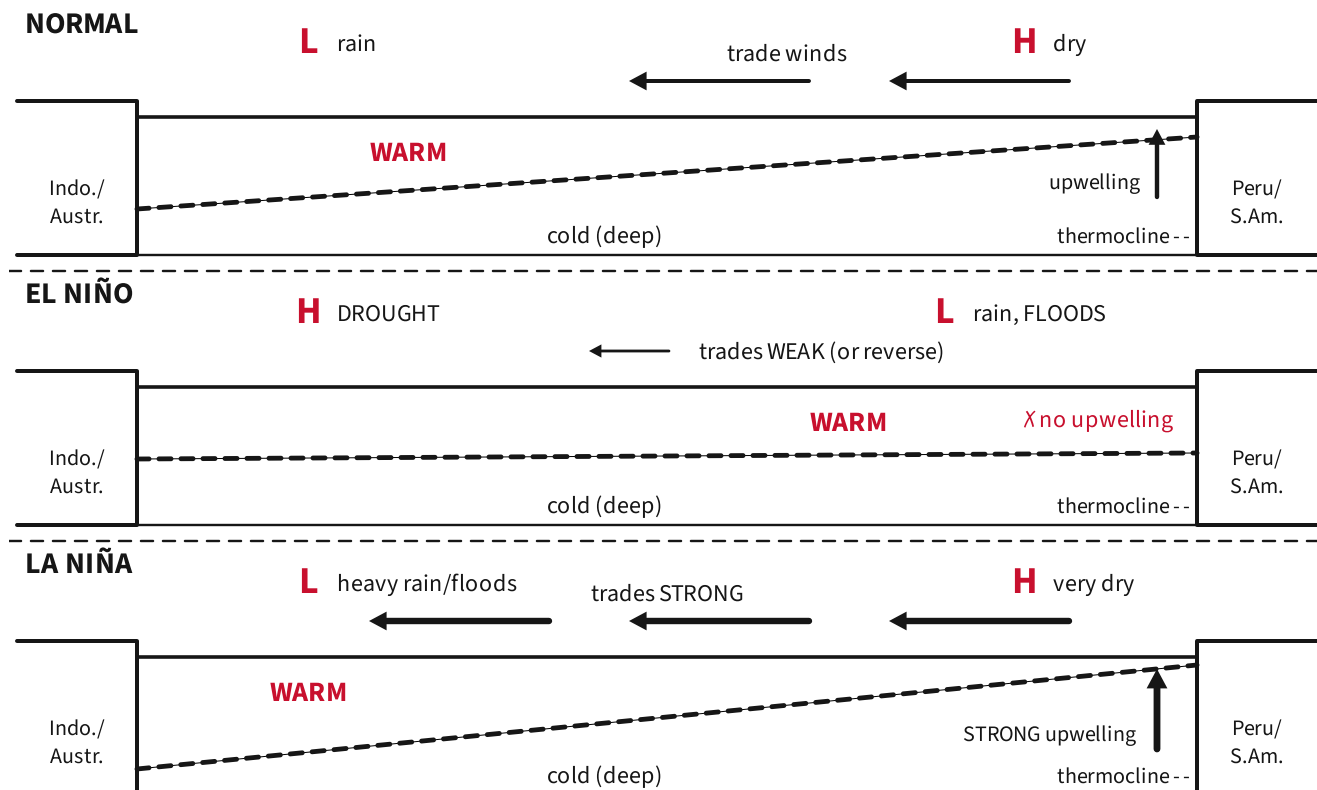
16

El Niño and La Niña — causes, effects, hurricanes ★

Review guide: Recognize and explain the causes/effects of El Niño/La Niña. How does this affect hurricane activity?

ANSWER — copy this

It's all about the **trade winds** over the equatorial Pacific. **Normal:** trades push warm water west. **El Niño:** **trades WEAKEN** → warm water slides back east. **La Niña:** **trades get STRONGER** → even more warm water piles up in the west.



How to read it: each panel is a slice across the equatorial Pacific (Australia left, Peru right). The dashed line is the thermocline. Watch it tilt: normal = tilted, El Niño = flat, La Niña = very tilted.

	NORMAL	EL NIÑO	LA NIÑA
Trade winds	blow east → west	weak (or reverse)	stronger
Warm water	piled up in the WEST	spreads EAST toward Peru	even more in the WEST
Peru / E Pacific	cold upwelling, great fishing, dry	NO upwelling → fish die; heavy rain + floods	strong upwelling, colder; very dry
Indonesia / Australia	rain	DROUGHT	heavy rain + floods
US winter	—	north warmer, south wetter	north colder, south drier
Atlantic hurricanes	—	FEWER	MORE
E Pacific hurricanes	—	more	fewer

KNOW THESE

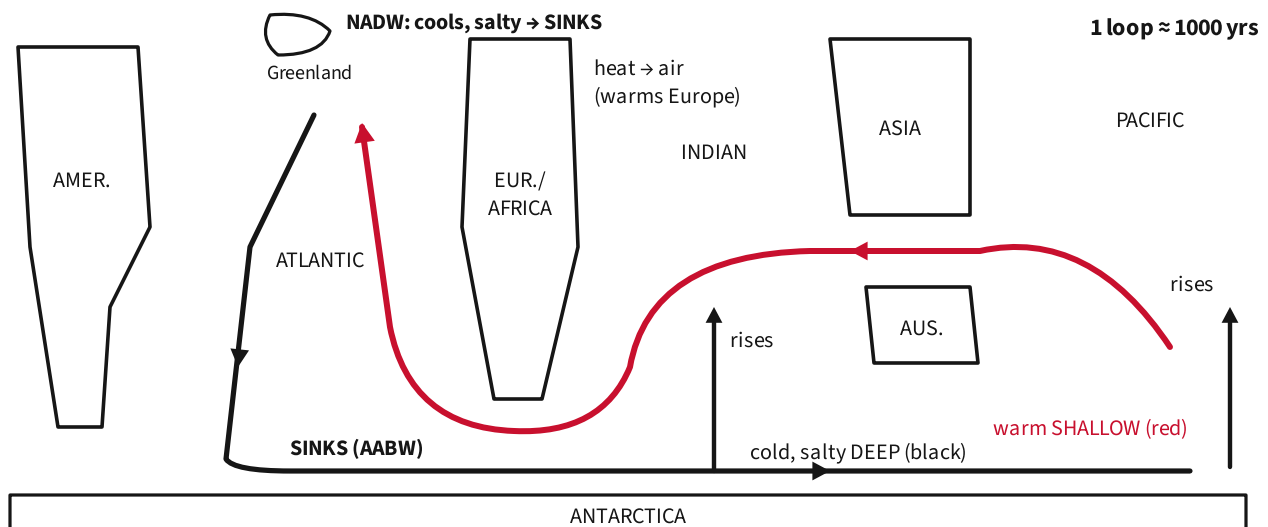
- **Why fewer Atlantic hurricanes in El Niño:** stronger vertical wind shear + trade winds and a more stable atmosphere. **La Niña** is the opposite: weaker shear + trades, less stable → more hurricanes.
- **ENSO** = El Niño–Southern Oscillation. The Southern Oscillation = the air-pressure seesaw between the west and east Pacific.
- **Anomaly** = difference from the average (e.g., an SST anomaly map shows where water is warmer or colder than normal).

17 How does deep water circulation affect climate? ★

Review guide: Explain how deep water circulation affects climate.

ANSWER — copy this

Near the poles, surface water becomes **cold and salty** (sea ice leaves salt behind), so it gets dense and **sinks**. That sinking drives the global **conveyor belt**: warm water flows toward the poles at the surface, sinks, and returns as cold deep water that slowly rises in the Indian and Pacific Oceans. **One loop takes ~1000 years**. It carries **heat** toward the poles and carries **O₂ and CO₂** into the deep ocean.



How to read it: red = warm, shallow return flow; black = cold, salty, deep flow. Water sinks at Greenland (NADW) and Antarctica (AABW) and rises in the Indian + Pacific.

THE CONVEYOR PATH

1. Warm surface water flows north in the Atlantic (Gulf Stream) and gives off heat to the air → keeps Europe mild.
2. Near Greenland it's now cold + salty → dense → **sinks** = NADW (North Atlantic Deep Water).
3. It flows south along the Atlantic bottom and joins **AABW** (Antarctic Bottom Water — the densest water, formed in the Weddell Sea).
4. It travels around Antarctica into the Indian and Pacific Oceans.
5. It slowly warms and **rises**, then returns at the surface to the Atlantic.

IF THE CONVEYOR SLOWS ★

- Melting ice + more rain → **fresher** (less salty) water → less dense → can't sink → conveyor slows → less heat carried north → the North Atlantic and Europe **cool**.
- It happened before: the **Younger Dryas** (~12,900–11,500 years ago). Meltwater from glacial **Lake Agassiz** poured out → the NH went back to near-glacial cold. It ended fast: Greenland warmed **10°C in about a decade**.
- Today: the Atlantic overturning (AMOC) slowed in the 20th century → a "cold blob" south of Greenland cooled while the rest of Earth warmed.
- Pacific deep water: weakly layered, sluggish, very uniform below 2000 m.